

Model-free Bayesian clustering and Multimodal posterior summarisation

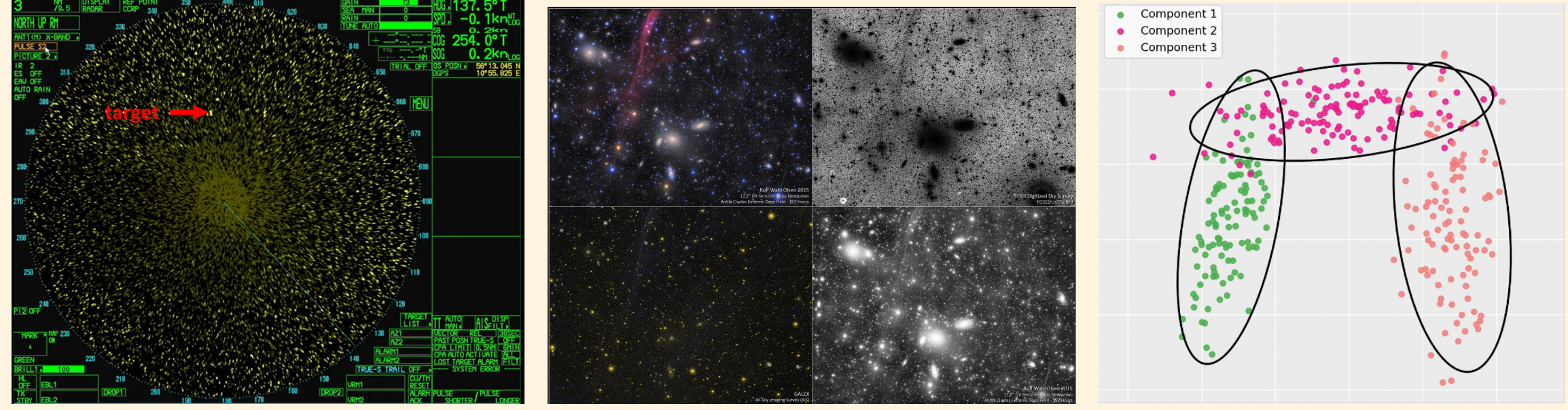
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1. Motivation and challenges

Many scientific clustering problems (astronomical surveys, object tracking) contain both structured signals and noise.

Problem

- Require good likelihood, priors, specifications of latent component parameters
- MCMC sampling can be expensive and posterior samples are hard to summarise
- Standard Bayesian mixture clustering may split one cluster into several mixture components.

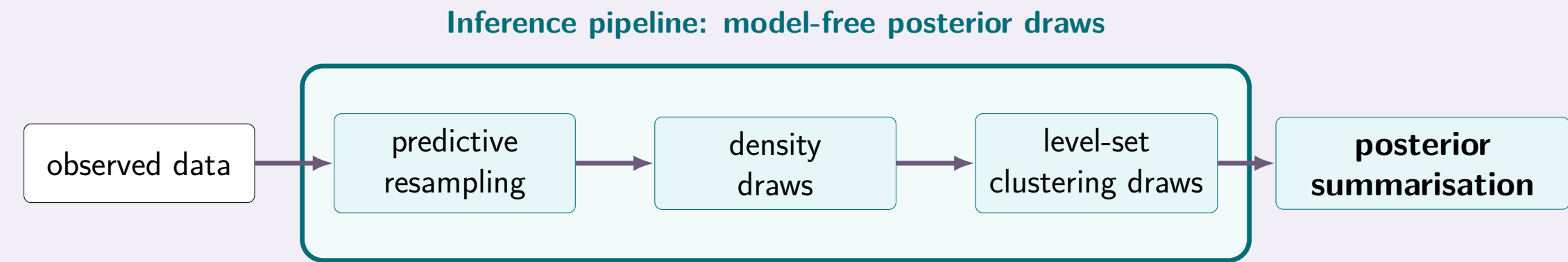


Main message

We develop a model-free noise-aware Bayesian clustering framework with posterior summarisation

Idea: Separate density learning from clustering: generate posterior density draws, convert each draw to a level-set sub-partition, then summarise the induced posterior on sub-partitions.

Output: Given the input data, the output is a compact, interpretable posterior approximation summarising multiple plausible clustering structures: $q_L = \sum_{\ell=1}^L w_\ell \delta_{\bar{c}_\ell}$, where each particle $\bar{c}_\ell$ represents a plausible noise-aware clustering mode and w_ℓ gives its posterior weight.



- Predictive resampling gives posterior density draws without committing to a fixed mixture allocation model.
- Each density draw is mapped to a level-set sub-partition: active clusters plus inactive/noise points.
- IA losses and WASABI compress the posterior into multiple weighted noise-aware clustering summaries.

1. Martingale Posterior on Clustering

Predictive Bayes setup: Uncertainty comes from the unobserved future $Y_{n+1:N} \mid y_{1:n}$.

Induced martingale posterior on sub-partitions. Let F_∞ be the limiting random predictive distribution, and define $\theta_\infty := \psi_{\delta, \lambda}(F_\infty)$. For measurable $B \subseteq \{0, 1, \dots\}^n$,

$$\Pi_\infty(\theta_\infty \in B \mid y_{1:n}) = \int \mathbf{1}\{\psi_{\delta, \lambda}(F_\infty) \in B\} d\Pi(F_\infty \mid y_{1:n}).$$

Posterior draws via predictive resampling

A coherent predictive sequence factorises as $p(y_{n+1:N} \mid y_{1:n}) = \prod_{i=n+1}^N p(y_i \mid y_{1:i-1})$.

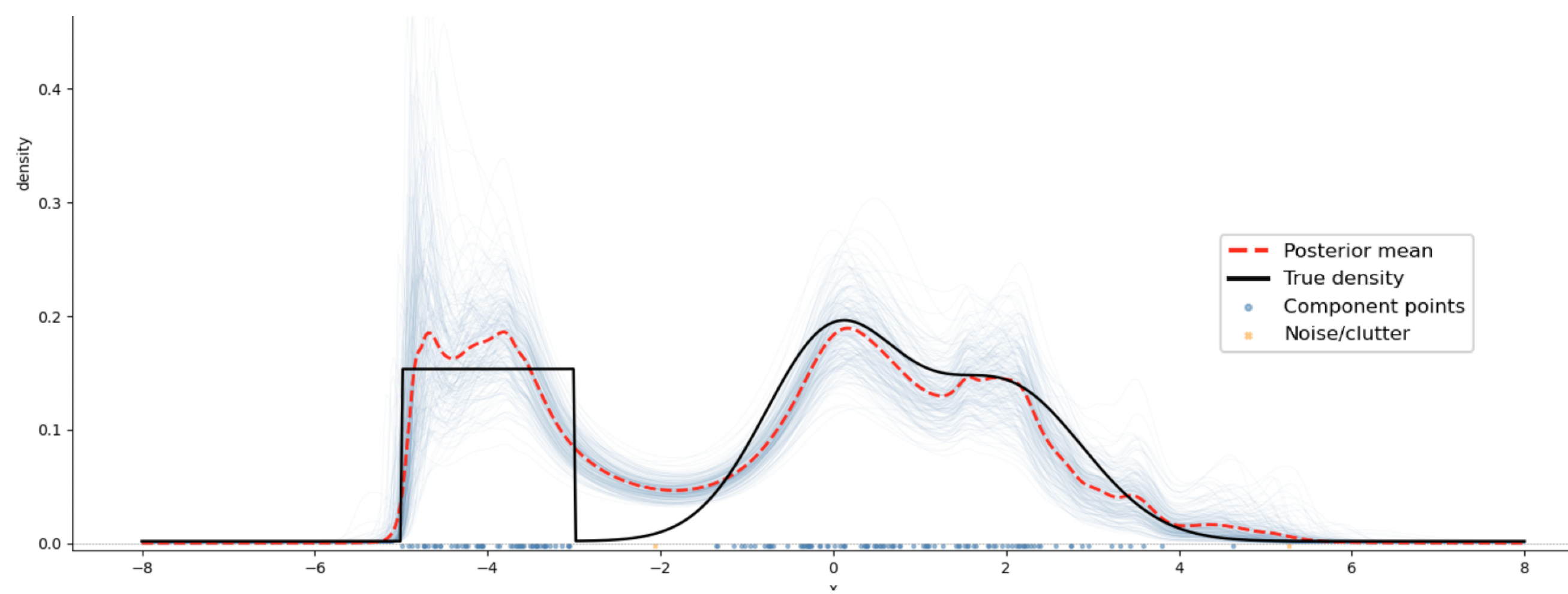
Starting from the fitted predictive after observing $y_{1:n}$, for each Monte Carlo run $j = 1, \dots, B$:



$\{\theta^{(j)}\}_{j=1}^B$ is a Monte Carlo approximation to $\Pi_\infty(\cdot \mid y_{1:n})$.

Eligiting the predictive sequence: Gaussian copula; Newton; TabMGP; etc.

Key point: Clustering uncertainty is induced by uncertainty in the predictive distribution



2. Density level-set sub-partitions

Level sets define clusters while naturally leaving low-density observations inactive.

Let f denote the unknown sampling density. For a level $\lambda > 0$,

$$S_\lambda(f) = \{x : f(x) \geq \lambda\}, \quad A_\lambda(f) = S_\lambda(f) \cap \{y_1, \dots, y_n\}.$$

The clustering target is a **sub-partition**: connected components of the level set, with observations outside the level set treated as inactive/noise.

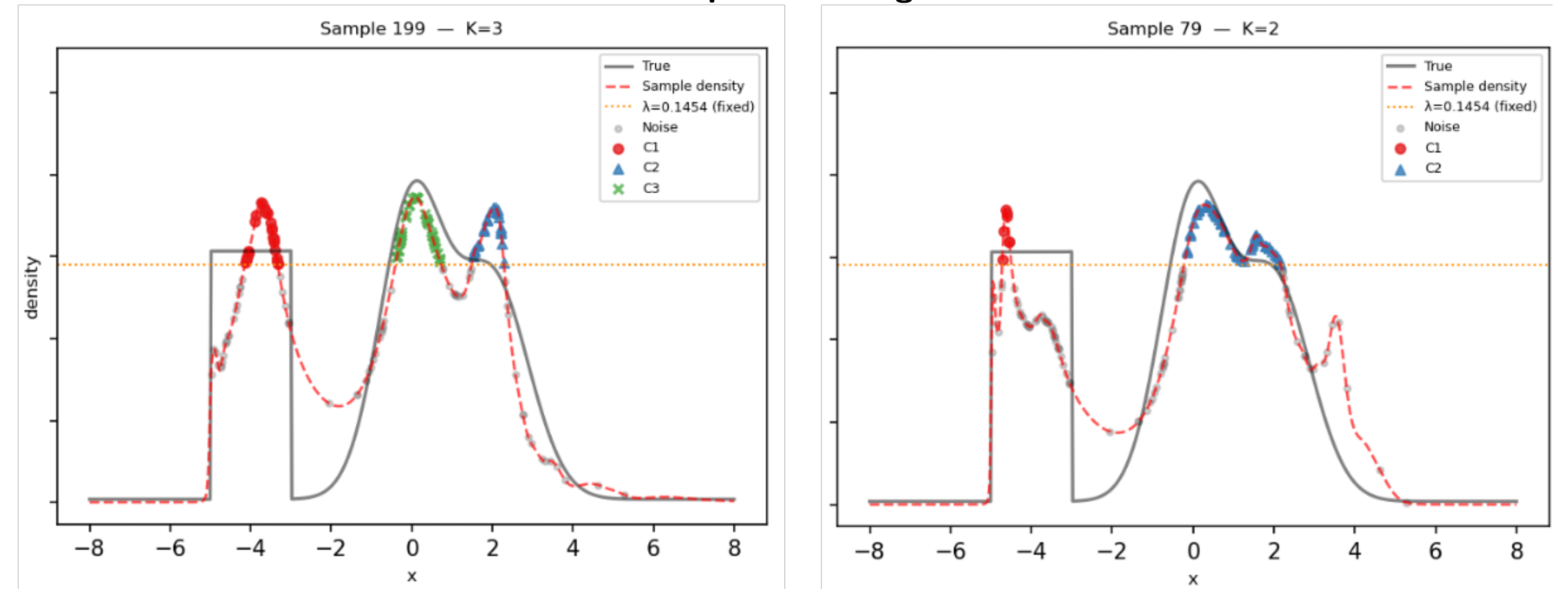
Sample Level-Set Graph Clustering

For $d \geq 2$, computing connected components of $L_\lambda(f)$ is nontrivial. We therefore use a discrete surrogate on the observed sample.

Three-step procedure:

- Threshold:** $A_\lambda(f) = \{i : f(y_i) \geq \lambda\}$. Points below λ are inactive/noise.
- Connect:** build a graph on $A_\lambda(f)$ using a δ -neighbourhood or reciprocal- k NN rule.
- Cluster:** connected components are clusters; inactive points receive label 0.

Two example clustering draws:



3. Multimode posterior summarisation—a compact, interpretable approximation to a large posterior sample of clusterings

Given posterior clustering draws $\theta^{(1)}, \dots, \theta^{(N_p)} \sim \pi(\theta \mid y)$,

Bayes estimator: minimise posterior expected loss:

$$\hat{\theta}^* = \arg \min_{\hat{\theta}} \mathbb{E}[L(\theta, \hat{\theta}) \mid y] \approx \arg \min_{\hat{\theta}} \frac{1}{N_p} \sum_{p=1}^{N_p} L(\theta^{(p)}, \hat{\theta}).$$

Loss function: inactive–active Variation of Information loss that penalises disagreement about active and inactive points $A = \{i : c_i \neq 0\}$ and $I = [n] \setminus A$.

$$VI_{IA}(\theta, \hat{\theta}) = \frac{\log_2(n)}{n} (m_{ai}[A \cap I'] + m_{ia}[I \cap A']) + VI(\rho_B, \hat{\rho}_B).$$

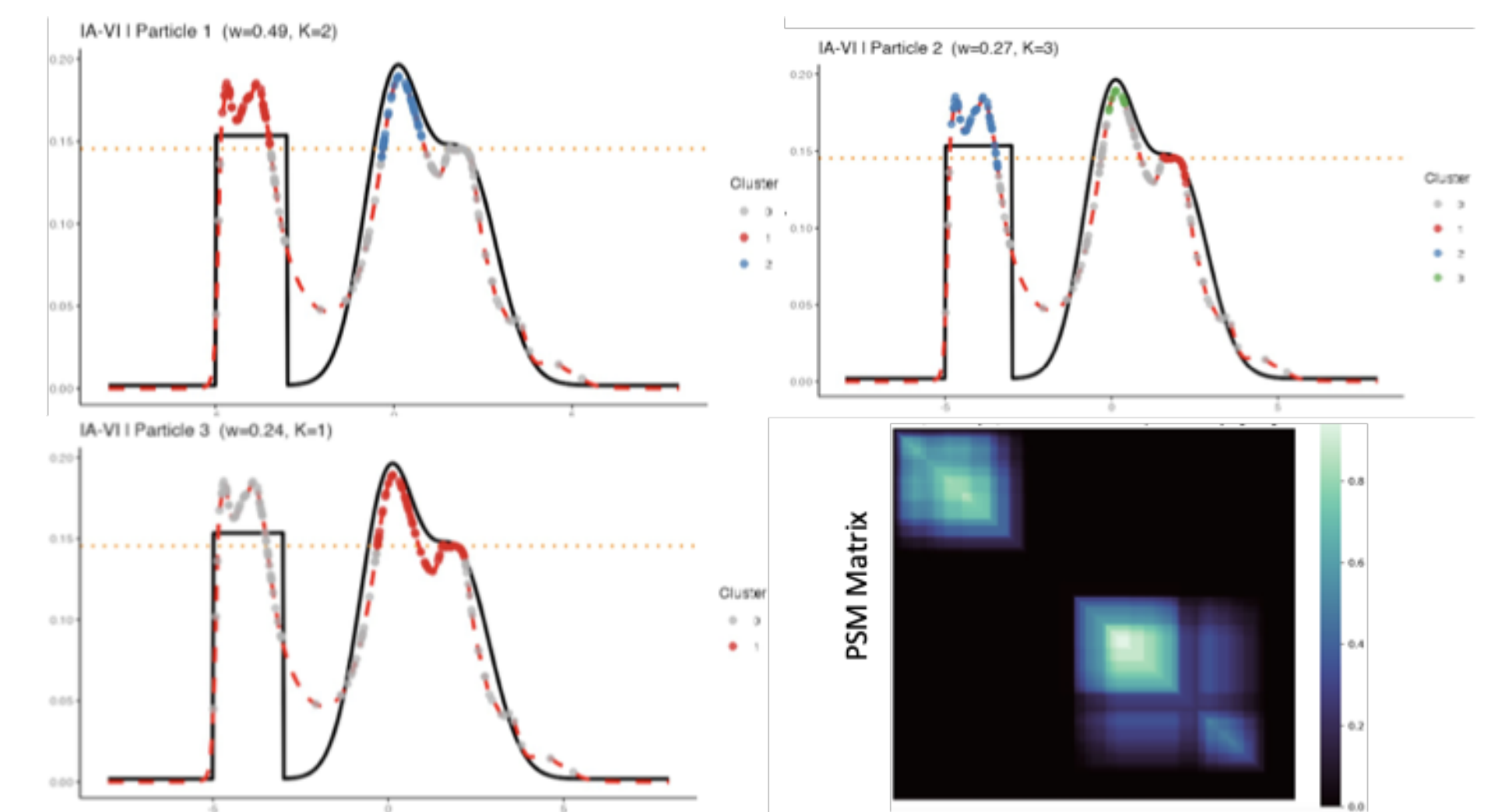
WASABI: approximate posterior by a small number of L weighted particles to capture different posterior modes $q_L(\theta) = \sum_{\ell=1}^L w_\ell \delta_{\theta_\ell}(\theta)$

Implementation: The particles are chosen to minimise a Wasserstein approximation:

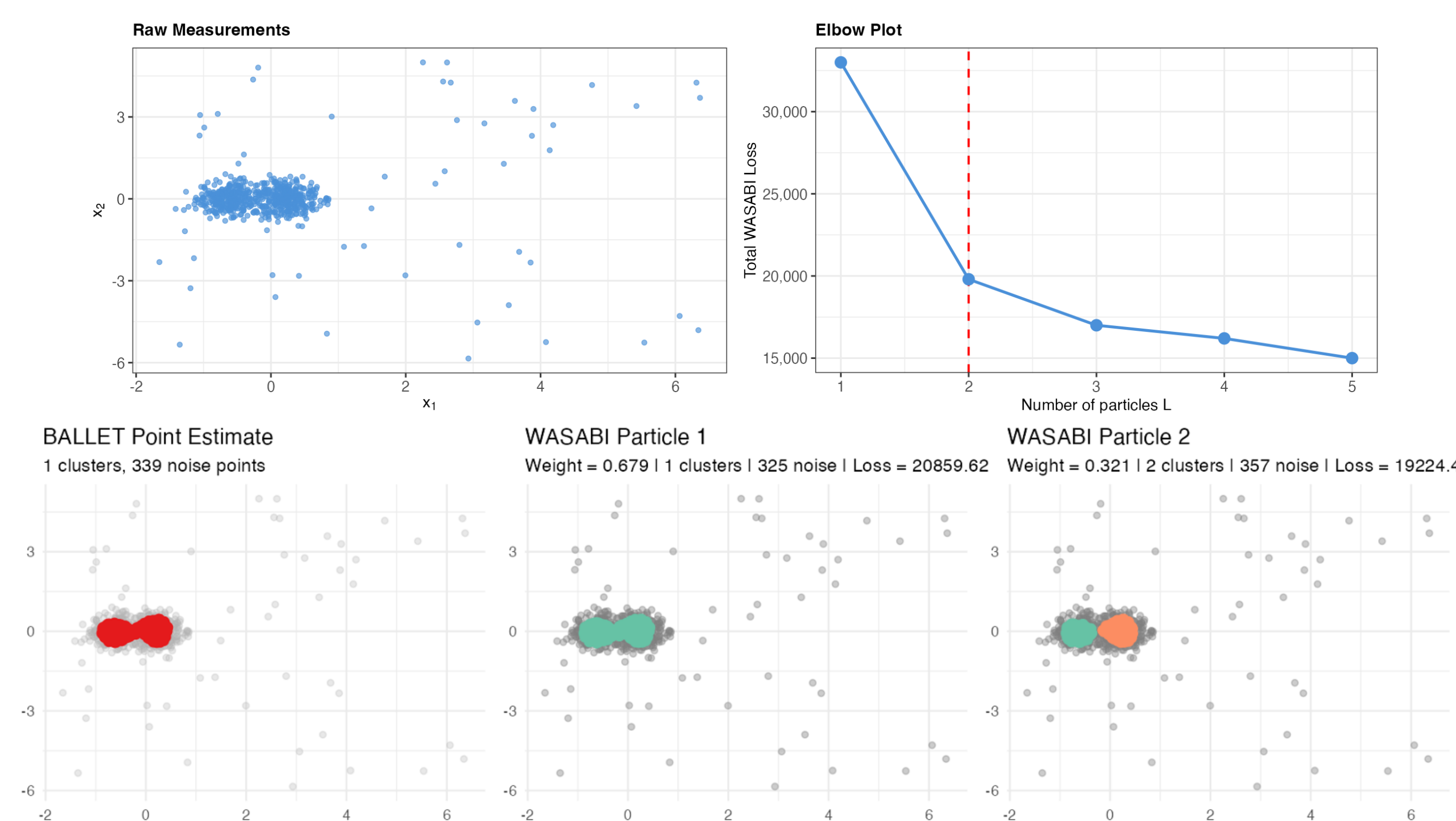
$$q_L^* = \arg \min_{q_L \in \mathcal{Q}_L} W_{L_{IA}}(\hat{\pi}, q_L), \quad \hat{\pi} = \frac{1}{T} \sum_{t=1}^T \delta_{\theta^{(t)}}.$$

This is implemented by a k -medoids-like procedure on the space of sub-partitions, iterate until convergence:

- N-update step:** Each sample $\theta^{(t)}$ is assigned to its closest particle θ_ℓ^*
 $N_\ell = \{\theta^{(t)} : \ell = \arg \min_{r=1, \dots, L} L_{IA}(\theta^{(t)}, \theta_r^*)\}$.
- Particle-search step:** Given a region of attraction N_ℓ ,
1. update particle $\theta_\ell^* = \arg \min_{\hat{\theta}} \frac{1}{|N_\ell|} \sum_{\theta^{(t)} \in N_\ell} L_{IA}(\theta^{(t)}, \hat{\theta})$.
2. Weight update: $w_\ell = \frac{|N_\ell|}{T}$.



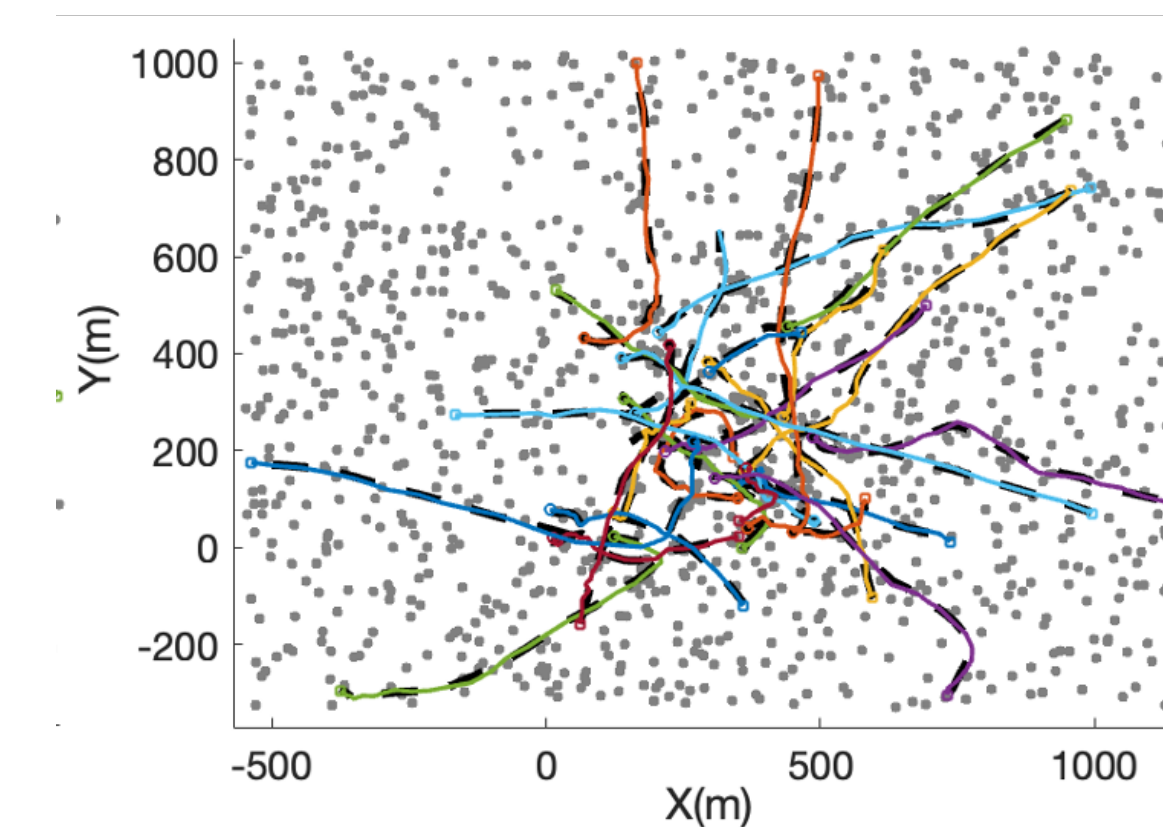
Experiment: bimodal clusters plus noise



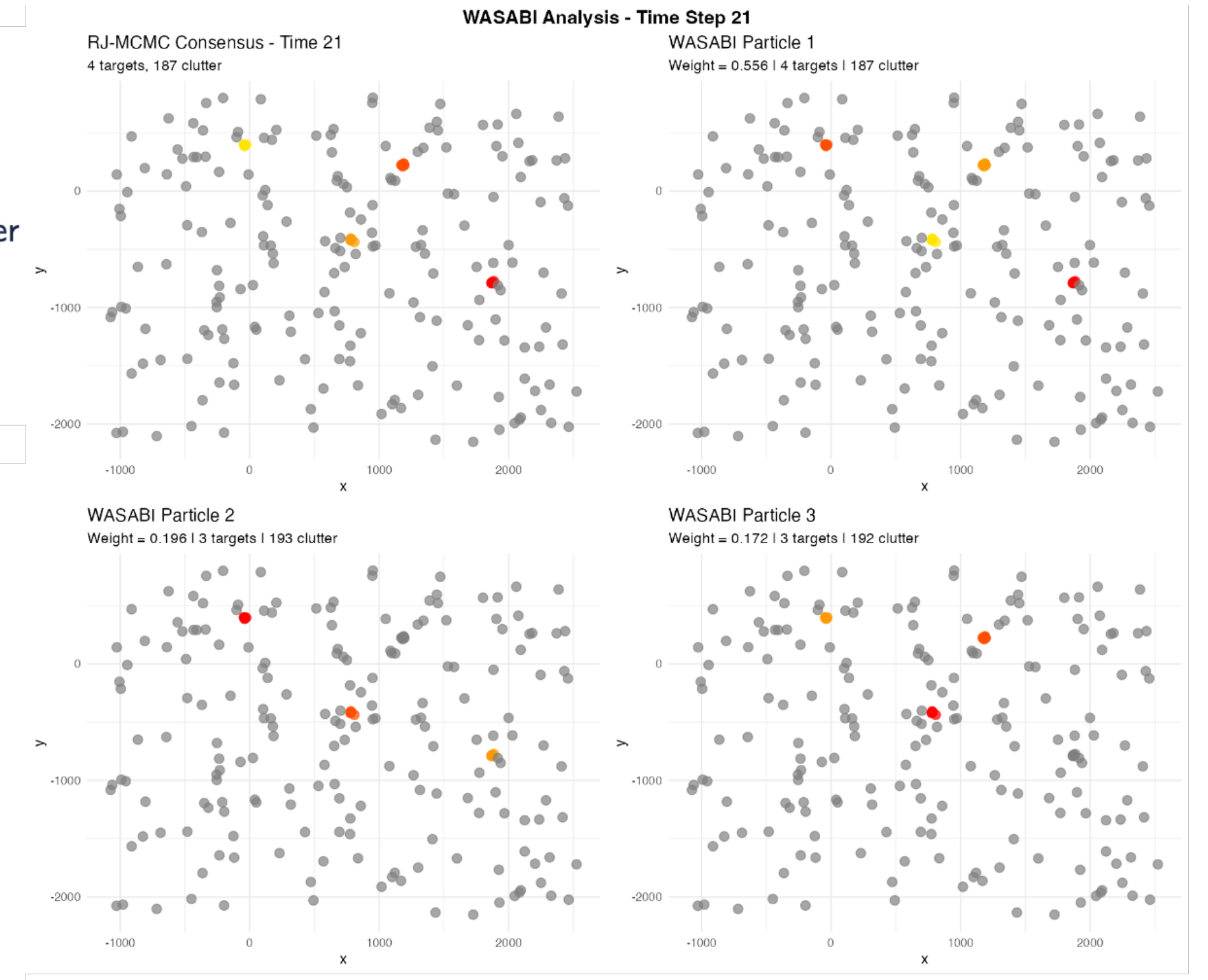
Application: multi-target tracking

Setting:

- Sensor returns include **true targets + clutter** (false alarms)
- Goal: identify true tracks, classify clutter as inactive
- Posterior draws of sub-partitions via sequential MCMC



Noise-aware summaries show stable tracks and ambiguous clutter associations.



Take-away message

- We propose a noise-aware posterior summarisation framework for Bayesian clustering.
- Inactive–active losses compare both sources of uncertainty: which observations are clustered, and how the active observations are grouped.
- WASABI with noise** compresses the posterior into a small set of weighted representative sub-partitions, capturing multimodality in both cluster structure and noise assignment.
- A **model-free pipeline** by using predictive resampling, combined with level-set graph clustering and WASABI-with-noise, delivering principled uncertainty quantification without specifying a likelihood or prior.

Reference and Contact info

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